DIVISOR CONDITIONS ON A SMOOTH RELATIVE CURVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 624-642 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 644-675, the complete author proof (30 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 12 of queue_pic.json (pb=1539, stmt 624-642, proof 644-675).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00PD: sources/stacks-pic/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AG8: sources/stacks-pic/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 062Y: sources/stacks-pic/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LS: sources/stacks-pic/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02KB: sources/stacks-pic/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01TX: sources/stacks-pic/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 056S: sources/stacks-pic/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a smooth morphism of schemes of relative dimension 1. Let $D \subset X$ be a closed subscheme. Consider the following conditions*

(1) *$D \to S$ is finite locally free,*
(2) *D is a relative effective Cartier divisor on X/S,*
(3) *$D \to S$ is locally quasi-finite, flat, and locally of finite presentation, and*
(4) *$D \to S$ is locally quasi-finite and flat.*

We always have the implications

$$(1) \Rightarrow (2) \Leftrightarrow (3) \Rightarrow (4)$$

If S is locally Noetherian, then the last arrow is an if and only if. If $X \to S$ is proper (and S arbitrary), then the first arrow is an if and only if.

Proof. Equivalence of (2) and (3). This follows from Divisors, Lemma 062Y if we can show the equivalence of (2) and (3) when S is the spectrum of a field k. Let $x \in X$ be a closed point. As X is smooth of relative dimension 1 over k and we see that $\mathcal{O}_{X,x}$ is a regular local ring of dimension 1 (see Varieties, Lemma 056S). Thus $\mathcal{O}_{X,x}$ is a discrete valuation ring (Algebra, Lemma 00PD) and hence a PID. It follows that every sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$ which is nonvanishing at all the generic points of X is invertible (Divisors, Lemma 0AG8). In other words, every closed subscheme of X which does not contain a generic point is an effective Cartier divisor. It follows that (2) and (3) are equivalent.

If S is Noetherian, then any locally quasi-finite morphism $D \to S$ is locally of finite presentation (Morphisms, Lemma 01TX), whence (3) is equivalent to (4).

If $X \to S$ is proper (and S is arbitrary), then $D \to S$ is proper as well. Since a proper locally quasi-finite morphism is finite (More on Morphisms, Lemma 02LS) and a finite, flat, and finitely presented morphism is finite locally free (Morphisms, Lemma 02KB), we see that (1) is equivalent to (2). □